Predefining an analysis plan

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Predefining an analysis plan before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Privacy and data governance

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Dataset and benchmark limitations

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Multiple comparisons

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Predefining an analysis plan trace and record the first step you can explain without help.

Use W2: Trace the worked case for Predefining an **analysis plan**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Predefining an analysis plan explanation against the supplied facts before discussing it.

Use the worked case for Predefining an **analysis plan**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Plan excludes malformed **input**s by a rule fixed beforehand. Three of15 are malformed; 9 of12 eligible pass. Report.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Predefining an analysis plan facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Predefining an analysis plan facts and keep your reasoning.

Plan includes all20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Predefining an analysis plan facts and keep your reasoning.

A report omits this qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Predefining an analysis plan facts and keep your reasoning.

Plan excludes malformed **input**s by a rule fixed beforehand. Three of15 are malformed; 9 of12 eligible pass. Report.

> **My reasoning and evidence**

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Predefining an analysis plan facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Predefining an analysis plan to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled predefining an **analysis plan evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of predefining an **analysis plan**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Predefining an analysis plan

> Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set. Explain the labelled step before checking the changed condition.

Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

AI ACADEMY / AI-7/21

Predefining an analysis plan

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Write the rule first

Record the outcome and justified unreadable-file exclusion.

2. Protect the reserved set

Do not choose exclusions from which cases each model fails.

3. Report deviations

If the plan changes, mark the change and give its reason.

CHANGE ONE THING

Exclude two readable cases because the new model fails them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If the plan changes, mark the change and give its reason.

1. Protect the reserved set

2. Write the rule first

3. Report **deviation**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report **deviation**s If the plan changes, mark the change and give its reason. An **analysis plan** limits flexibility after outcomes are known.

Necessary changes can be transparent, but they should not be silently presented as the original plan.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Exclude two readable cases because the new model fails them. Which conclusion is supported by the supplied facts?

1. That outcome-driven exclusion violates the recorded rule.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That outcome-driven exclusion

violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

Repair a missing building block

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

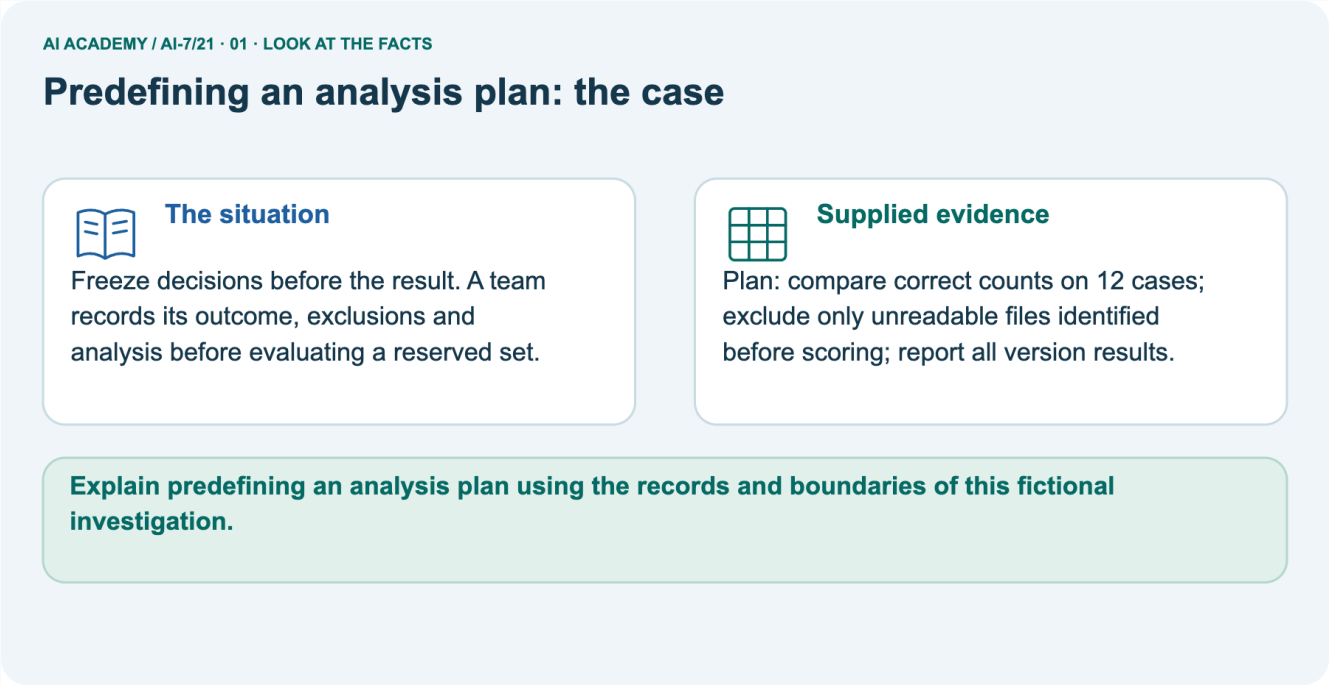
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Predefining an analysis plan picture from its labels, then record a relevant limit.

01. Predefining an analysis plan: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

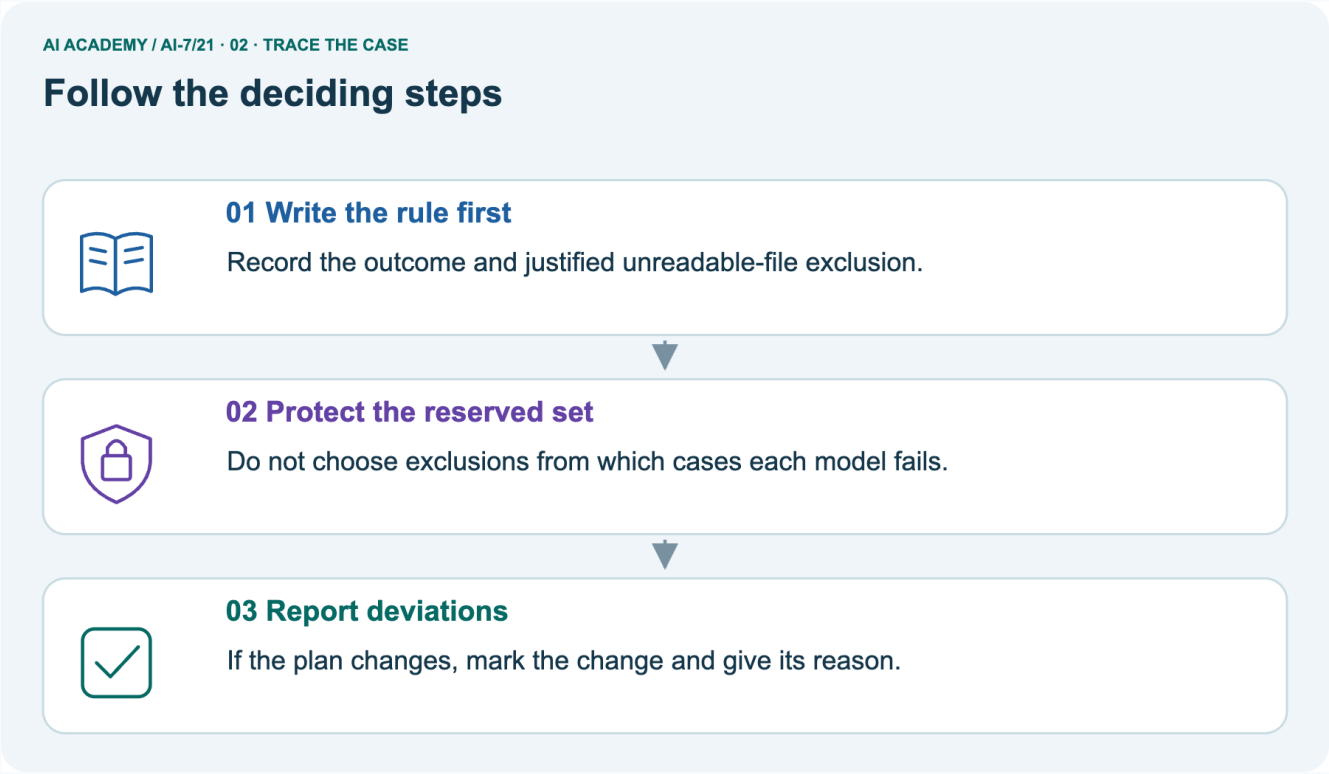
Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Supplied facts: Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



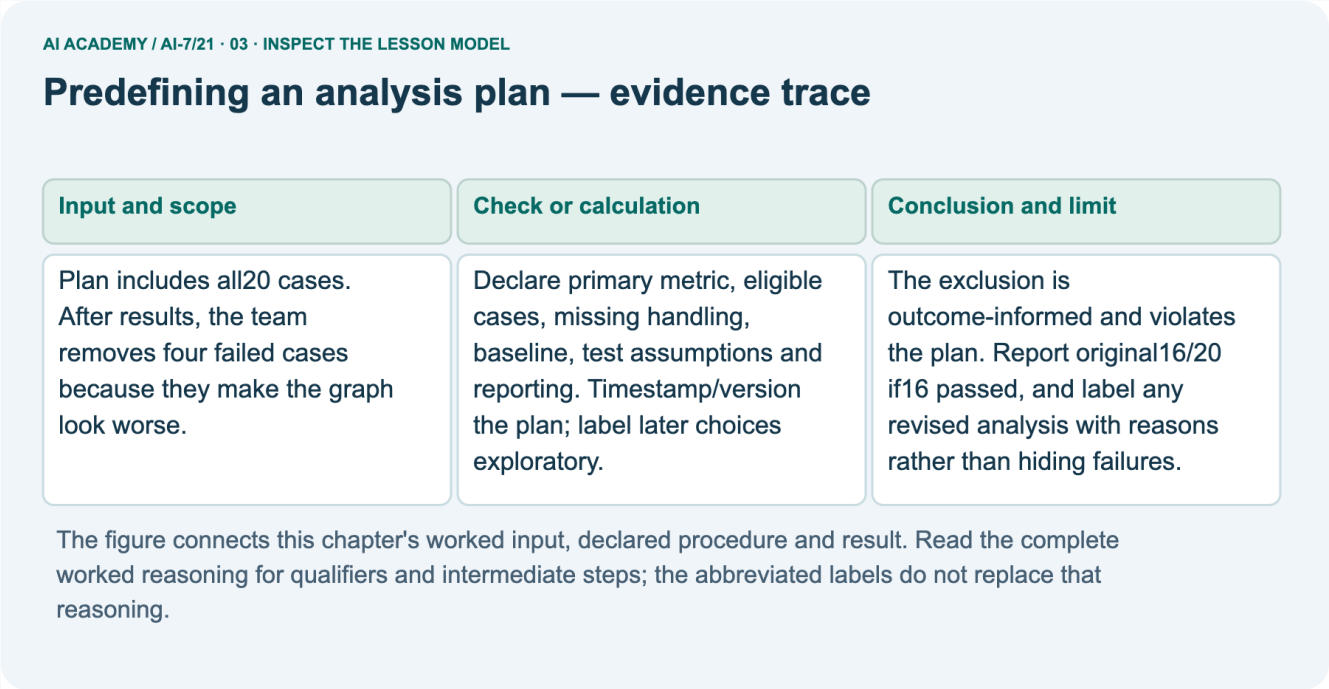
Follow the labelled picture with the matching explanation. Open the vector diagram

Write the rule first: Record the outcome and justified unreadable-file exclusion. → Protect the reserved set: Do not choose exclusions from which cases each model fails. → Report **deviation**s: If the plan changes, mark the change and give its reason.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Predefining an analysis plan — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Predefining an **analysis plan** — **evidence** trace. Plan includes all20 cases.

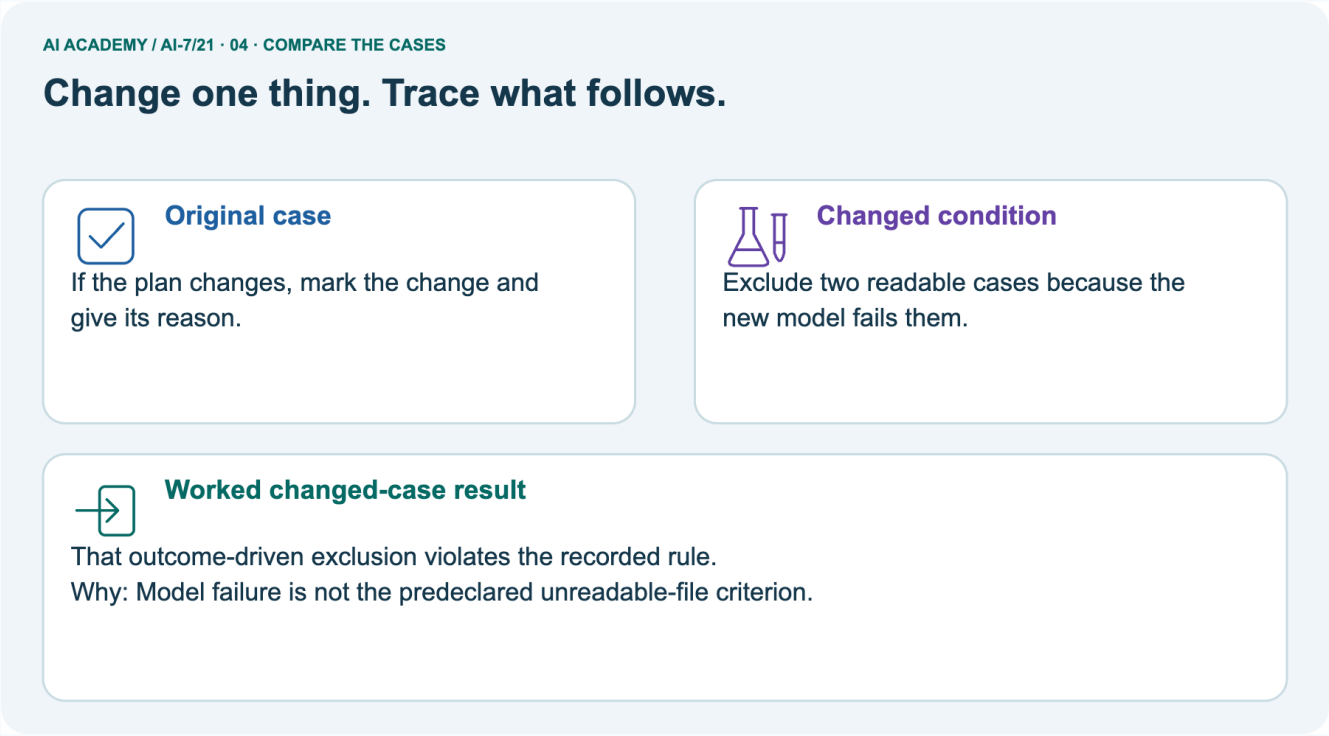
After results, the team removes four failed cases because they make the graph look worse. | Declare primary metric, eligible cases, missing handling, baseline, test assumptions and reporting.

Timestamp/version the plan; label later choices exploratory. | The exclusion is outcome-informed and violates the plan. Report original16/20 if16 passed, and label any revised analysis with reasons rather than hiding failures.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

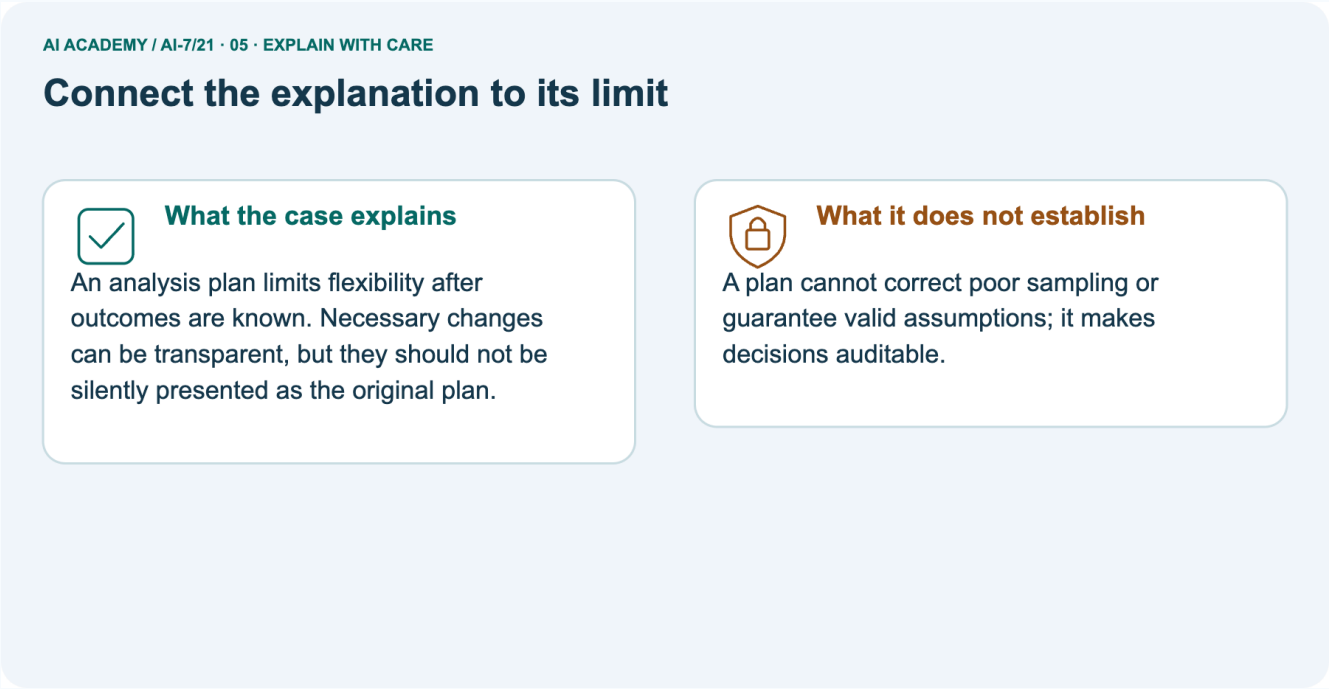
Original case: If the plan changes, mark the change and give its reason. Changed condition: Exclude two readable cases because the new model fails them.

Expected result: That outcome-driven exclusion violates the recorded rule. Reason: Model failure is not the predeclared unreadable-file criterion.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

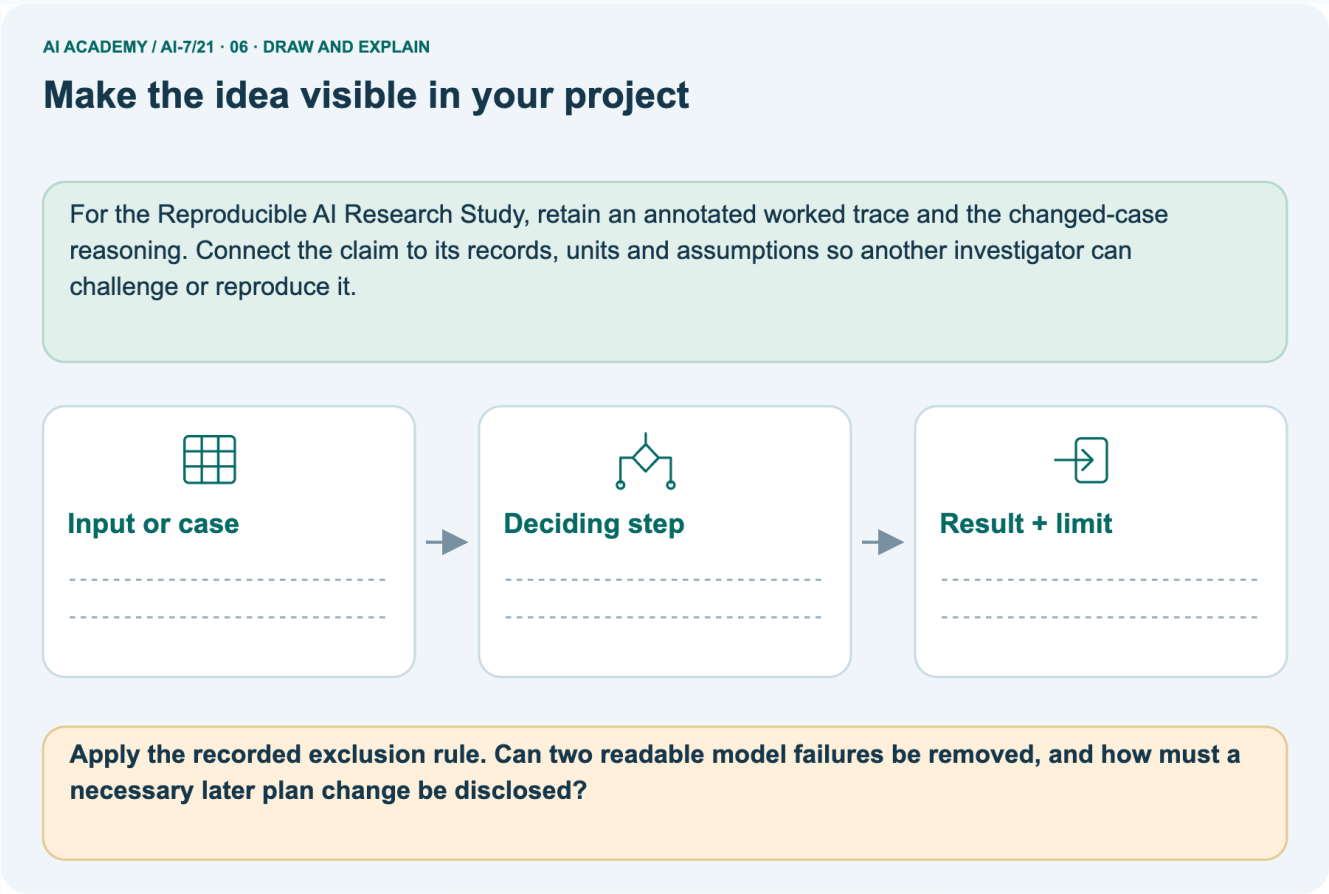
An **analysis plan** limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

Limit: A plan cannot correct poor sampling or guarantee valid assumptions; it makes decisions auditable.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the recorded **exclusion rule**. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.